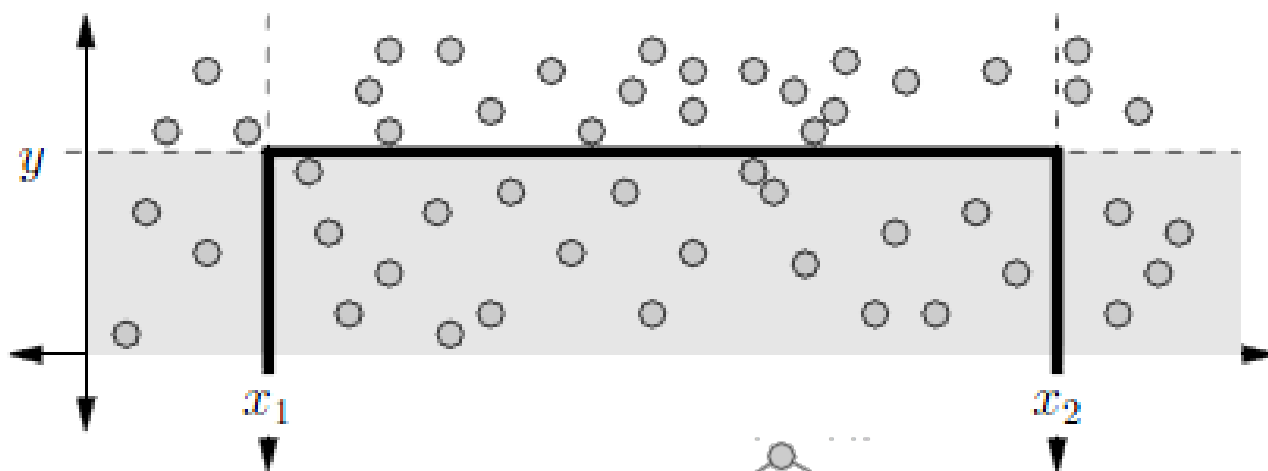


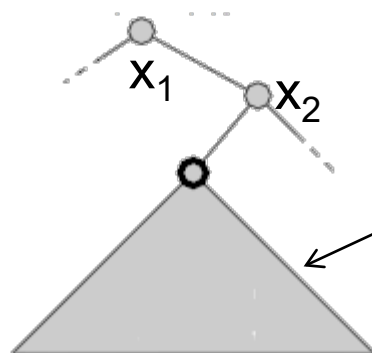
Five Nice Ideas...

5. Augment Tree to Help with Queries

- Answer “3-sided” range emptiness queries...



Augment each node with
min y in its subtree...

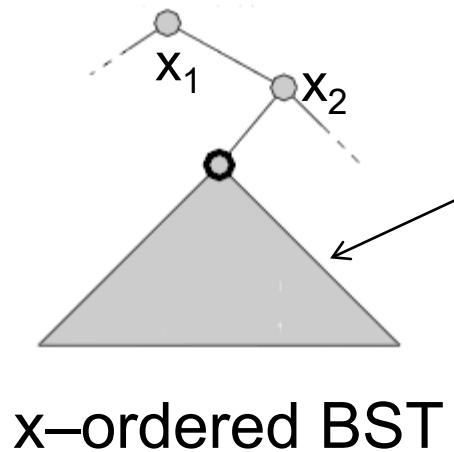
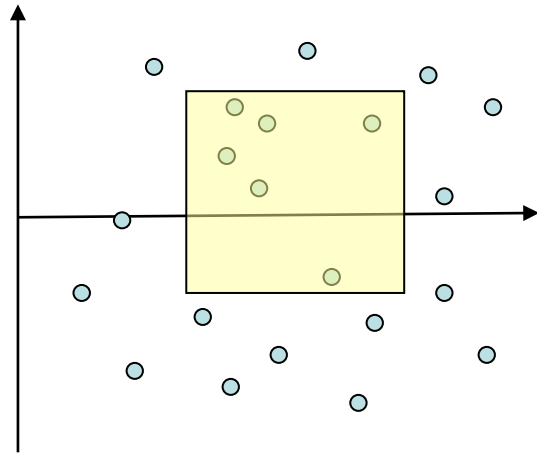


Points with x
coordinates in
range (x_1, x_2)

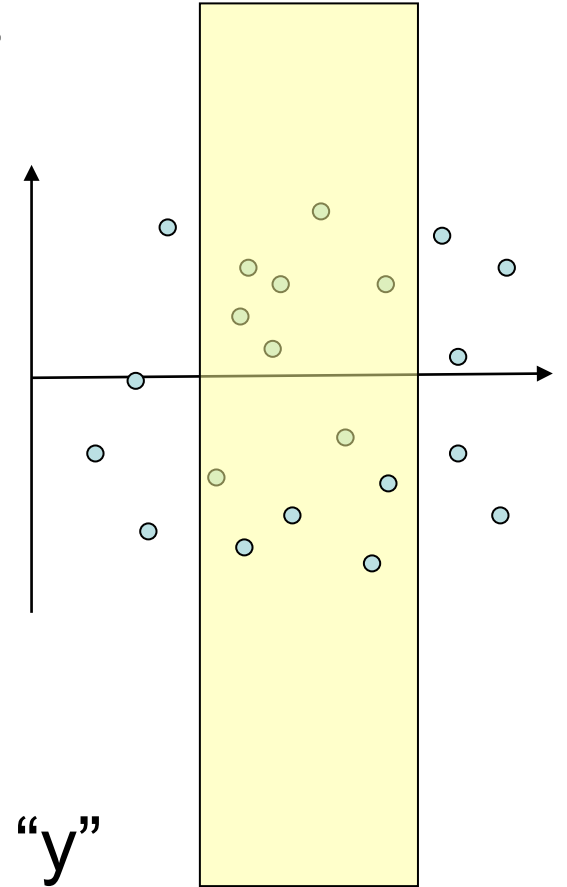
Five Nice Ideas...

5. Augment Tree to Help with Queries

- What about “4-sided” queries? Can augmentations help here?



Points with x
coordinates in
range (x_1, x_2)



- Need to filter elements of this subtree matching the “y” part of range...